

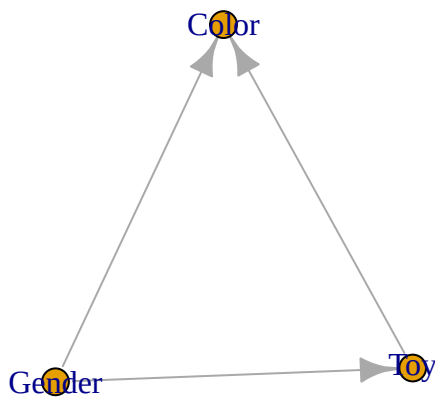
Worksheet 2

Exercise 1: Children's Toys

Imagine a study with children in which the influence of gender (boy, girl) on the color (black, blue, pink) and type of toy (car and dolls) is examined. Girls tend to play with dolls rather than cars and prefer the color pink. Furthermore, it is generally known that the color and type of toy depend on each other: cars are more likely to be black and blue, while dolls are more likely to be dressed in pink.

- a) Draw the causal diagram for the three variables **gender**, **color** and **toy**.

```
library(gRbase)
# List all nodes with their parents
graph.toys <- dag(c("Gender"),
                  c("Color", "Gender", "Toy"),
                  c("Toy", "Gender"))
# Draw graph
plot(graph.toys)
```



- b) The data.frame **toys** in the file **toys.rda** contains data from this study. Build a Bayesian network using the causal diagram from a) and the data.

```
load("toys.rda")

library(gRain)
# combine graph and data to a Bayesian network
gn <- grain(graph.toys, data = toys)
```

- c) Calculate the two conditional probability:
i.e. $P(\text{Toy} = \text{car} \mid \text{Gender} = \text{girl})$ and $P(\text{Toy} = \text{car} \mid \text{Gender} = \text{boy})$.
Describe in your own words how these probabilities should be interpreted.

```
# Option 1: from data
# P(Toy = car | Gender = girl)
toys_girls <- toys[toys$Gender == "girl",]
```

```
sum(toys_girls == "car")/nrow(toys_girls)
```

```
[1] 0.3915344
```

```
# P(Toy = car | Gender = boy)
toys_boys <- toys[toys$Gender == "boy",]
sum(toys_boys == "car")/nrow(toys_boys)
```

```
[1] 0.8931624
```

```
# Option 2: from the Bayesian network:
querygrain(gn, nodes=c("Toy", "Gender"), type="conditional")
```

	Gender	
Toy	boy	girl
car	0.8931624	0.3915344
doll	0.1068376	0.6084656

The conditional probabilities are

$P(\text{Toy} = \text{car} \mid \text{Gender} = \text{girl}) = 0.3915$ and $P(\text{Toy} = \text{car} \mid \text{Gender} = \text{boy}) = 0.8931$.

For the conditional probability $P(\text{Toy} = \text{car} \mid \text{Gender} = \text{girl})$, we consider only the group of girls. The conditional probability describes the proportion of children within this subgroup of girls that have car as favorite toy.

d) Calculate the two joint probabilities

$P(\text{Toy} = \text{car}, \text{Gender} = \text{girl})$ and $P(\text{Toy} = \text{car}, \text{Gender} = \text{boy})$.

What is the difference compared to the conditional probabilities in b)?

```
# P(Toy = car, Gender = girl)
n_car_girl <- sum(toys$Toy == "car" & toys$Gender == "girl")
n_car_girl/nrow(toys)
```

```
[1] 0.1749409
```

```
# P(Toy = car, Gender = boy)
n_car_boy <- sum(toys$Toy == "car" & toys$Gender == "boy")
n_car_boy/nrow(toys)
```

```
[1] 0.4940898
```

```
# or from the Bayesian network:
querygrain(gn, nodes=c("Toy", "Gender"), type="joint")
```

	Toy	
Gender	car	doll
boy	0.4940898	0.05910165
girl	0.1749409	0.27186761

In contrast to question b), we consider all children, i.e. we calculate the probability among all children of being a girl and preferring a car. In b), we calculated the probability preferring a car only among the girls.

e) Describe the meaning of the probability $P(\text{Toy} = \text{car} \mid \text{do}(\text{Gender} = \text{girl}))$? Why is it impossible to observe data for this?

$P(\text{Toy} = \text{car} \mid \text{do}(\text{Gender} = \text{girl}))$ describes the probability that a child chooses a car as a toy if we actively set the child's gender to girl through an intervention. In reality, we cannot randomly or

experimentally change a child's gender. We can only observe which toys children choose as favorite given their natural gender. The `do()` operator, however, means that all other causes of gender are removed and gender is actively set. This is not feasible in a real study.

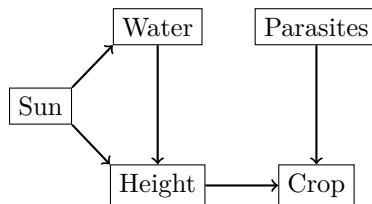
f) Which paths transport causal effect from gender on toy? Justify your answer.

Causal effects can only be transported along the direction of arrows. There is only one **directed** paths from Gender to Toy:

Gender $\rightarrow$ Toy.

Exercise 2: Apple tree

Have a look at the example from the lecture on the crop of apple trees.



We assume the following discrete values for the five random variables:

- Sun $\in \{\text{little, much}\}$
- Water $\in \{\text{little, much}\}$
- Height $\in \{\text{little, much}\}$
- Crop $\in \{\text{little, much}\}$
- Parasites $\in \{\text{yes, no}\}$

The following probabilities are given:

- $P(\text{Sun} = \text{much}) = 0.7$
- $P(\text{Water} = \text{much} \mid \text{Sun} = \text{much}) = 0.05$ and
 $P(\text{Water} = \text{much} \mid \text{Sun} = \text{little}) = 0.65$.
- $P(\text{Height} = \text{much} \mid \text{Sun} = \text{much}, \text{Water} = \text{much}) = 0.4$,
 $P(\text{Height} = \text{much} \mid \text{Sun} = \text{little}, \text{Water} = \text{much}) = 0.45$,
 $P(\text{Height} = \text{much} \mid \text{Sun} = \text{much}, \text{Water} = \text{little}) = 0.55$,
 $P(\text{Height} = \text{much} \mid \text{Sun} = \text{little}, \text{Water} = \text{little}) = 0.25$.
- $P(\text{Parasites} = \text{yes}) = 0.18$
- $P(\text{Crop} = \text{high} \mid \text{Height} = \text{much}, \text{Parasites} = \text{yes}) = 0.67$,
 $P(\text{Crop} = \text{high} \mid \text{Height} = \text{little}, \text{Parasites} = \text{yes}) = 0.01$,
 $P(\text{Crop} = \text{high} \mid \text{Height} = \text{much}, \text{Parasites} = \text{no}) = 0.9$,
 $P(\text{Crop} = \text{high} \mid \text{Height} = \text{little}, \text{Parasites} = \text{no}) = 0.42$.

a) Build a Bayesian network with R using the given conditional probabilities. Visualize the resulting causal diagram.

```

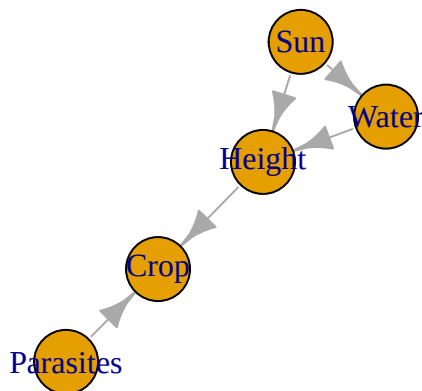
a <- cptable("Sun", values = c(0.7, 0.3), levels = c("much", "little"))
b <- cptable(c("Water", "Sun"), values = c(0.05, 0.95, 0.65, 0.35),
             levels = c("much", "little"))
c <- cptable(c("Height", "Sun", "Water"),
             values = c(0.4, 0.6, 0.45, 0.55,
                       0.55, 0.45, 0.25, 0.75),

```

```

        levels = c("much","little"))
d <- cptable(c("Crop","Height","Parasites"),
            values = c(0.67, 0.33, 0.01, 0.99,
                      0.9, 0.1, 0.42, 0.58),
            levels = c("high","low"))
e <- cptable(c("Parasites"), values = c(0.18, 0.82), levels = c("yes","no"))
# Building the complete Bayesian Network
plist.apple <- compileCPT(list(a, b, c, d, e))
bn.apple <- grain(plist.apple, compile = FALSE)
plot(bn.apple)

```



- b) Calculate the marginal probabilities of a high crop yield, $P(\text{Crop} = \text{high})$, and a low crop yield, $P(\text{Crop} = \text{low})$, using R.

```
querygrain(bn.apple, nodes="Crop", type="marginal")
```

```

$Crop
Crop
      high      low
0.5991975 0.4008025

```

- c) What is the probability of a high crop yield given that a tree has parasites?

```
querygrain(bn.apple, nodes=c("Crop","Parasites"), type="conditional")
```

```

      Parasites
Crop      yes    no
high 0.335875 0.657
low  0.664125 0.343

```

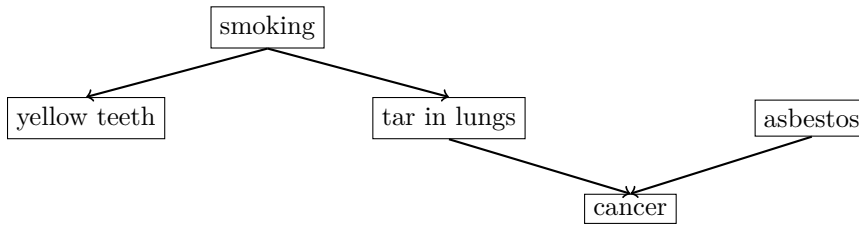
$P(\text{Crop} = \text{high} \mid \text{Parasites} = \text{yes}) = 0.336$.

- d) Are Height and Parasites independent given the Crop?

No, there is a collider between Height and Parasites. Crop is the collision node. Therefore, Height and Parasites are conditionally dependent given Crop.

Exercise 3: Lung cancer

Have a look at the following causal graphical model about lung cancer.



Which of the following independence statements hold?

1. **yellow teeth** is independent of **tar in lungs** conditioning on **smoking**.
2. **tar in lungs** is independent of **asbestos**.
3. **tar in lungs** is independent of **asbestos** conditioning on **cancer**.
4. **smoking** is independent of **cancer** conditioning on **tar in lungs**.
5. **yellow teeth** is independent of **tar in lungs**.

1. Yes, because there is a fork (**yellow teeth** $\leftarrow$ **smoking** $\rightarrow$ **tar in lungs**). If we condition on the common cause **smoking** then the two direct effects become independent.
2. Yes, because there is a collider between the two nodes. **tar in lungs** and **asbestos** are therefore independent causes.
3. No, because this is a collider (**tar in lungs** $\rightarrow$ **cancer** $\leftarrow$ **asbestos**). If we condition on the collider node (i.e. **cancer**), then the two causes become dependent.
4. Yes, because there is a chain (**smoking** $\rightarrow$ **tar in lungs** $\rightarrow$ **cancer**) and we condition on the middle node **tar in lungs**.
5. No, because there is a fork between the two nodes **yellow teeth** and **tar in lungs**. The two variables share a common cause (**smoking**), which we do not condition on, and they are therefore dependent.